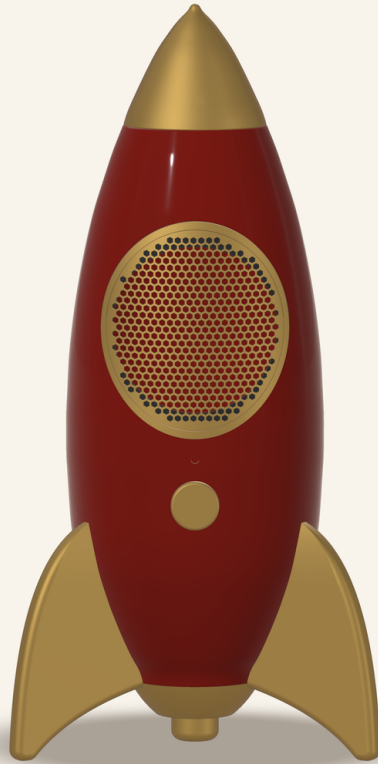




BATHSHEVA LONDON



Atelier

Prototype build manual

OXBLOOD · LOOKS-LIKE PROTOTYPE · SEPTEMBER 2026

INTRODUCTION

Before you begin

From the box of printed resin parts to a finished prototype, ready to photograph and shake down.

Atelier is built from eight printed pieces (the fins print once and go on three times), finished by hand in two colours and glued together in a set order. Nothing here is difficult, but each stage needs patience: thin coats of paint, time to cure, and a dry fit before any glue. Each step ends with a checkpoint; don't move on until it is met.

TIMING

Weekend one

Inspect and dry-fit, prepare the surfaces, prime, and paint the colours.

The week between

Let the 2K clear coat harden. It feels dry within hours but needs days to cure fully.

Weekend two

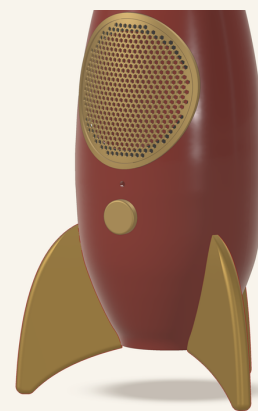
Assemble, weigh it down, and check it stands and photographs well.

SAFETY

- Spray outdoors or somewhere very well ventilated, and wear the respirator whenever you spray.
- Wear a dust mask or the respirator, and nitrile gloves, when sanding resin: its dust shouldn't be breathed in or left on skin.
- The 2K clear coat contains isocyanates. Spray it outdoors only, wear a proper A2P3 organic vapour and particulate respirator (a dust mask is not enough), and make sure nobody else, and no pets, are nearby while spraying or while it flashes off.
- Wear nitrile gloves when mixing or applying epoxy.
- Keep sprays and epoxy away from children and pets, and let fumes clear before bringing parts indoors.

HOW TO USE THIS MANUAL

Work through the seven steps in order. Each part's colour and treatment is listed on the next page; keep it open while you paint. If something goes wrong, turn to Troubleshooting at the back: almost every mistake can be sanded out and repainted.



Fins on, dry-fitted.

INVENTORY

Parts and supplies

PART	QTY	FINISH
■ Body	1	Oxblood gloss, 2K clear coat
■ Nose cone	1	Metallic gold
■ Fins	3	Metallic gold
■ Grille	1	Metallic gold; recess floor behind it masked and painted matt black
■ Bezel	1	Metallic gold
■ Knob	1	Metallic gold; decorative on the prototype, doesn't turn
■ Collar	1	Metallic gold
■ Foot	1	Metallic gold

SUPPLIES

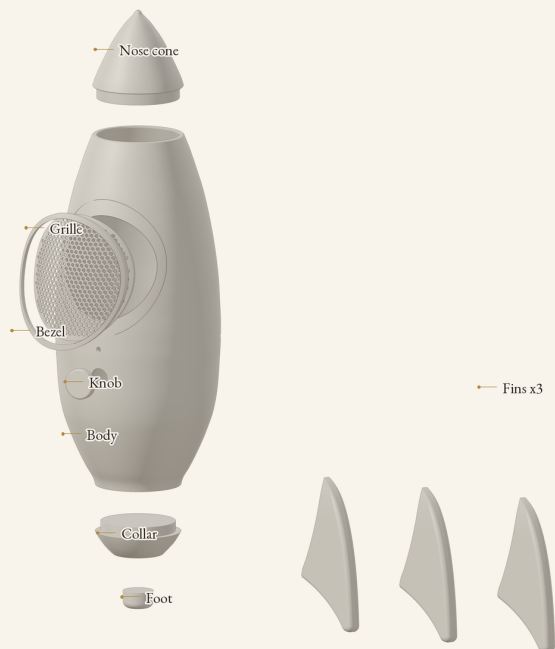
- U-POL Isolate Acid #8 etch primer, or U-POL Custom Grey primer-filler (proposed, primer)
- Oxblood/deep red basecoat to match #8A1C15, colour-matched by a U-POL or automotive refinish supplier (proposed, colour, confirm on a swatch, Step 04)
- U-POL 2K HS Clearcoat, aerosol or 2K gun mix (proposed, clear; contains isocyanates, see Safety)
- Metallic gold spray, e.g. Rust-Oleum Universal Metallic "Gold" (proposed, for the trim parts, left uncoated)
- Matt black acrylic and a small brush, for the grille recess floor
- Wet-and-dry paper, P240 to P1200
- Tamiya masking tape, 10 mm
- Araldite Rapid epoxy, cocktail sticks
- M4 x 12 mm self-tapping screws, 6 (proposed, for the fins, into their 3.3 mm pilot holes; epoxy alone is the fallback, see Step 06)
- Steel shot or fishing weights, about 500 g, for the internal weight (proposed)
- A2P3 respirator (for the 2K clear coat), dust mask, nitrile gloves
- Craft knife, blu-tack
- A spare printed swatch or offcut, for the Step 04 paint test

Every specific product above is proposed, not confirmed: the primer, colour and clear are one compatible system (Step 04 still opens with a swatch test before the full spray-out), but the exact colour match and the clear's cure schedule need checking against a real can before you commit to it.

STEP 01

Inspect and dry-fit

Check every part and fit them together without glue. Paint adds thickness, so tight fits only get tighter.



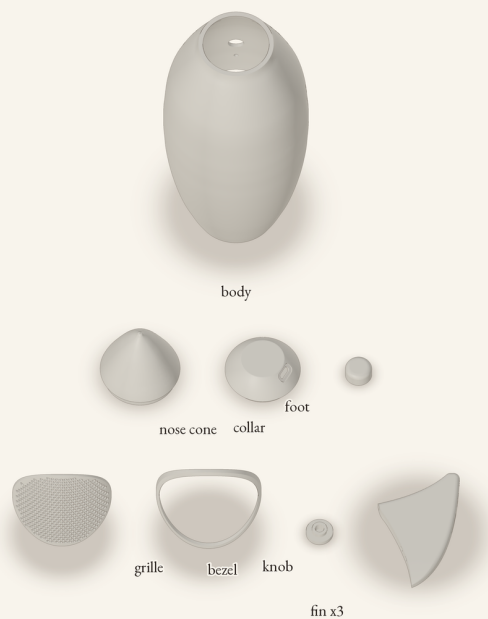
- 01 Tick off every part against the inventory. Look for cracks, warping or resin flash.
- 02 Stand the collar on the foot, then hold the body over the collar's spigot: it should sit straight, without forcing.
- 03 Press the knob's boss (its own moulded peg, 10 mm across, 3 mm long) into the body's knob hole (10.8 mm across, 0.4 mm clearance all round). It should push in with light resistance and no rocking.
- 04 Hold the grille in its recess, then the bezel over its edge. Both should sit flush, without a gap or a lip proud of the body.
- 05 Dry-fit each fin against the body: the flat root should sit flush against the hull with its pilot holes (3.3 mm) aligned to the body's own.
- 06 Stand the nose cone on the body's top rim: it should sit centred and level.
- 07 Stand the whole dry stack where it would live, and judge size and proportions.

STEP 02

Prepare the surfaces

Clean, smooth surfaces are what make the paint look expensive.

- 01 Wash all parts in warm soapy water, rinse, and leave to dry fully.
- 02 Trim any support marks with a craft knife, cutting away from yourself.
- 03 Put on a dust mask or the respirator, and gloves. Sand visible surfaces with P240 to flatten marks, then P400 to smooth.
- 04 Wipe the dust off with a barely damp cloth and let dry.



Every printed part, as it arrives in plain white resin.

HANDLE WITH CARE

- **Grille:** the honeycomb webs are fine and can snap; only remove support marks from the flat back, never press on the front face.
- **Bezel:** its edge is thin all the way round; sand it flat on a board rather than freehand, so it doesn't go out of round.
- **Fins:** the trailing edge tapers to almost nothing; support it in your hand while you sand so it doesn't flex or crack.

READY TO MOVE ON WHEN

every part is clean, dry and dust-free, with no support marks left on the surfaces that show.

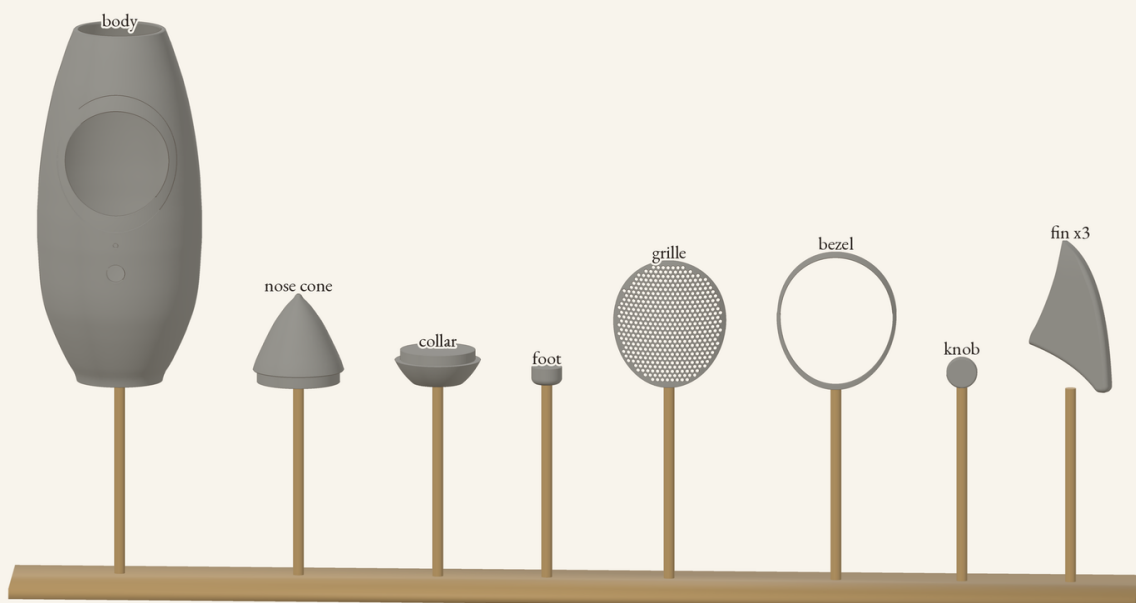
STEP 03

Prime

Primer fills fine layer lines and shows every flaw. This is where the final finish is really made.

- 01 Mount parts on a tube, bottle or cocktail sticks with blu-tack, so you can reach all round without touching.
- 02 Choose a dry, still day, around 15 to 25°C. Shake the primer for two to three minutes.
- 03 Spray light coats from 25 to 30 cm, starting and finishing each pass off the part.
- 04 Let it dry, then sand with P400, wearing the dust mask. Repeat until the surfaces look smooth under a bright light.
- 05 Prime every part, body included.

Drying times. Typically touch-dry in 15 to 30 minutes, ready for the next coat after about 30 minutes, and ready to sand after at least an hour. Brands vary, so check the can.



Every part mounted for priming.

READY TO MOVE ON WHEN

every part is an even grey, with no layer lines, pits or bare spots under a bright light.

STEP 04

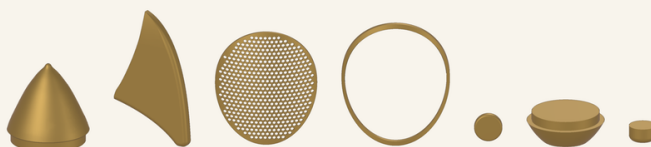
Paint

Two colours, so there's little masking between them. Tape only the surfaces that will be glued, and test the system on a swatch before you commit the real parts.

COLOUR	PARTS
■ Oxblood gloss	body
■ Metallic gold	nose cone, fin x3, grille, bezel, knob, collar, foot
■ Matt black	grille recess floor (on the body, masked)



Oxblood gloss



Metallic gold



Matt black

SWATCH TEST FIRST

Spray the primer, oxblood colour and 2K clear on a spare printed offcut in that order, exactly as you plan to spray the body, and let it cure fully (see Step 05's timings). Some colours wrinkle or lift under a 2K clear; if the swatch stays smooth and hard, carry on.

MASK BEFORE PAINTING

- The body's cone-joint rim and collar spigot
- The grille and bezel's back edges, where they seat in the recess
- The back of the knob's boss
- Each fin's flat root, where it glues to the hull

SPRAYING

- 01 Three or four light coats, 25 to 30 cm away.
- 02 On the grille, spray at an angle into the honeycomb so the inner walls are covered.
- 03 If dust or a run appears, let it dry, sand with P1200 and respray lightly.
- 04 Once the body's oxblood coat is dry, mask the honeycomb opening and hand-brush the recess floor matt black; wipe the raised surround clean before it dries.

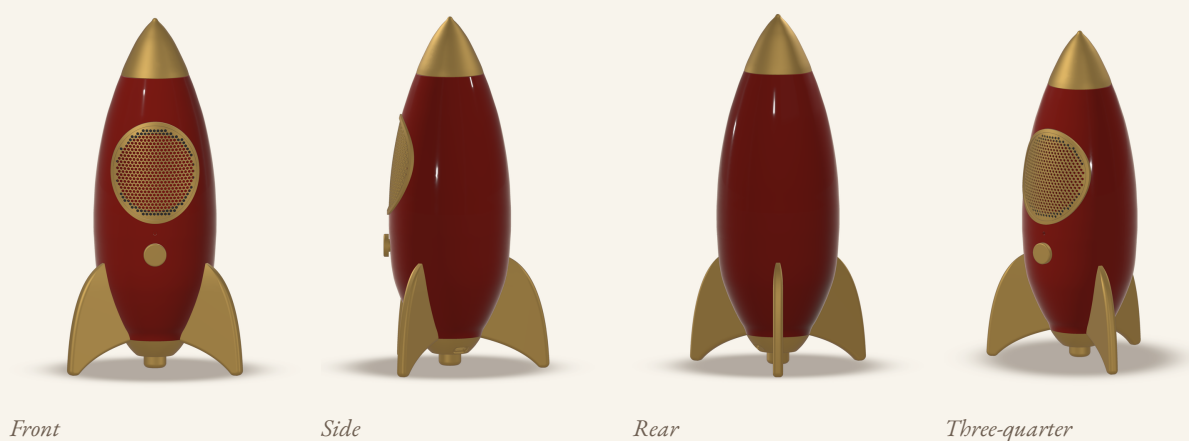
Drying times. Typically touch-dry in 15 to 30 minutes, with the next coat after 15 to 30 minutes. Check the can.

The 2K clear coat is next, in Step 05; this step's checkpoint comes after it.

STEP 04, CONTINUED

Colour reference

The body and grille recess floor take their colour here; the gold trim parts are covered in Step 06 as they're glued on, so mask and paint them alongside the body even though they're fitted later.



READY TO MOVE ON WHEN

every part is evenly coloured with no bare patches or runs, and the masking tape is off.

STEP 05

2K clear coat and cure

The gold trim is left uncoated (a clear coat dulls metallic paint), but the body's oxblood gloss is clear-coated for the lacquer look.

- 01 Wait at least 24 hours after the oxblood colour coat.
- 02 2K clear contains isocyanates: spray outdoors, wear the A2P3 respirator, and make sure nobody else is nearby (see Safety, Before you begin).
- 03 Mix the 2K clear exactly to the pack ratio, in a well-ventilated spot, and use it within its pot life.
- 04 Spray two light coats on the body only.
- 05 Leave it warm, dry and dust-free for two to three days. Paint feels dry long before 2K clear is fully hard.
- 06 Optional: once fully hard, polish the body with car polishing compound.

Drying times. 2K clear is typically touch-dry in 30 to 60 minutes, recoatable within an hour or after 24 hours (check the can, many 2K products have a recoat window either side of which you must key the surface), and fully cured in 2 to 3 days at room temperature.

READY TO MOVE ON WHEN

the body's gloss is even and hard: a fingernail pressed on a hidden spot leaves no mark, and the gold trim parts are unclouded.

STEP 06

Assemble

Araldite Rapid gives about five minutes to position a joint. Mix small amounts, one joint at a time, and hold or tape each joint as it sets.

- Wait about 20 to 30 minutes before handling a glued joint or starting the next one.
- Epoxy reaches full strength the next day: leave the finished prototype overnight before lifting it by the nose cone or testing its stability.
- Dry-fit each joint just before you glue it, and wipe off any squeeze-out with a cocktail stick before it sets.

ORDER

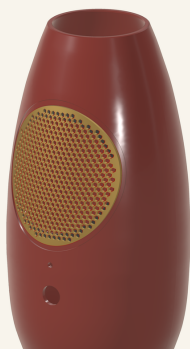
Front details first, then the fins, then the weight, then seal the base and cap the nose: grille, bezel, knob, fins, weight, collar, foot, nose cone. The weight has to go in before the collar, while the body's underside is still open.

THE KNOB'S OWN BOSS, NOT A DOWEL

The knob is moulded with its own mounting peg on the back: a 10 mm boss, 3 mm long, sized for the body's own 10.8 mm knob hole (0.4 mm clearance all round). That boss is what mounts the knob; no separate dowel or sleeve is needed. (The knob also has its own 6 mm blind bore, for a production encoder shaft; on the prototype it's just left empty behind the boss.)

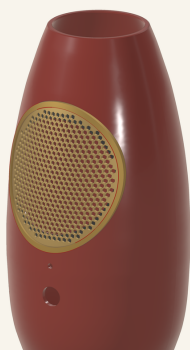
STEP 06, ASSEMBLY, 1 OF 2

Assembly order



01 Grille

The grille recess floor was painted matt black in Step 04; check it's fully dry, then glue the grille into the recess.



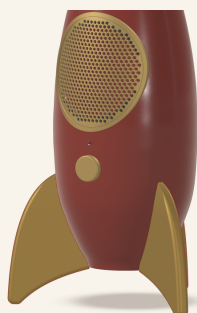
02 Bezel

Over the grille's edge, into the same recess.



03 Knob

Glue the knob's 10 mm boss into the body's own knob hole. On the prototype it is decorative and doesn't turn.

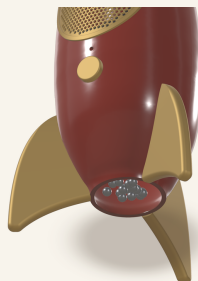


04 Fins

Align each fin's pilot holes (3.3 mm) with the body's, then fix with M4 self-tapping screws from inside, or epoxy alone if you'd rather not drill.

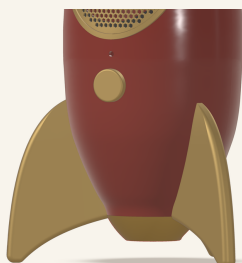
STEP 06, ASSEMBLY, 2 OF 2

Assembly, continued



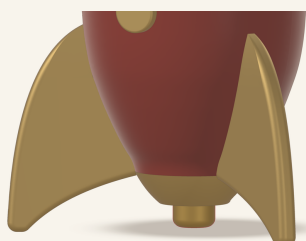
05 Weight

Glue about 500 g of steel shot low inside the body, through the open bottom, clear of the knob boss and fin screws.



06 Collar

Friction-fits onto the body's underside spigot (0.2 mm clearance); glue it in place, sealing the weight inside.



07 Foot

Glue centred under the collar.



08 Nose cone

Glue onto the body's top rim, centred.

READY TO MOVE ON WHEN

the glue has set overnight, the fins are firm with no flex, and the prototype stands level on the foot without rocking.

STEP 07

Stability check and photograph

The model's own numbers describe the production part, not this hand-glued, hand-weighted resin one, so this step is a physical test, not a rendered prediction.

TIP TEST

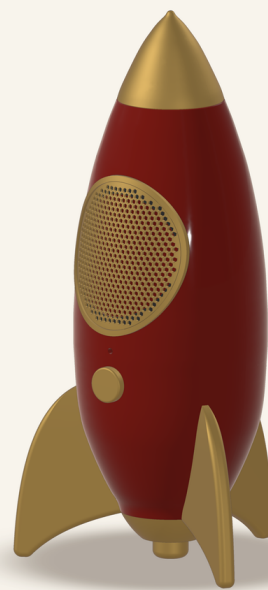
- 01 Stand the finished prototype on a flat, level surface, resting on its three fin tips and the centre foot.
- 02 Without the steel shot fitted, nudge it gently from the side at about knob height and see how little it takes to tip.
- 03 With the steel shot glued in (as built), repeat the same nudge and compare.
- 04 Try it on a soft surface (a rug or cushion) too, and note whether it's noticeably less stable.

WHAT GOOD LOOKS LIKE

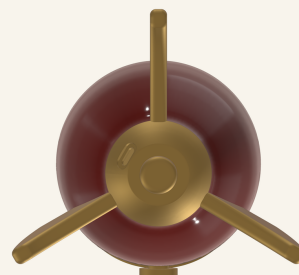
- It resists a firm, deliberate nudge and only tips under a hard knock.
- It stands level on all three fins and the foot, without rocking.
- It feels reassuringly solid to pick up, not hollow.

PHOTOGRAPHS

- Two sets: plain backdrop, and one styled shelf scene.
- Front, side, three-quarter, rear, grille close-up, underside.
- Natural daylight if you can; note the light source either way.



Three-quarter, for reference.



Underside: fins and foot take the weight.

READY TO MOVE ON WHEN

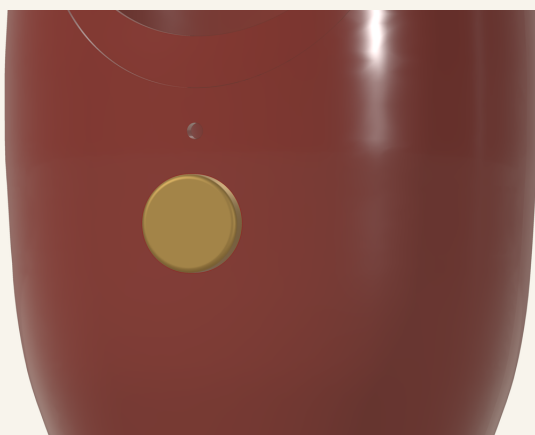
it passes the tip test both weighted and unweighted, the test record is filled in, and both photo sets are taken.

REFERENCE

Troubleshooting

Almost every problem is recoverable. Let paint harden fully before sanding or recoating.

PROBLEM	FIX
Knob boss won't seat, or rocks	Sand the boss lightly with P400 and test again; it should press into the body's 10.8 mm hole with light resistance, not force.
Fin sits proud, or rocks	Check the pilot holes are aligned before driving the screws; a fin that's still proud after gluing can be sanded flush at its root and touched up.
Paint run or sag	Let it harden, sand flat with P1200 and spray a light coat over it.
Rough, bumpy gloss (orange peel)	Sand gently with P1200 and polish, or add another light 2K clear coat. Next time, lighter coats.
Dust specks	Once hard, sand the speck with P1200 and respray lightly on a still day.
2K clear wrinkles, lifts or stays soft	Let it harden as far as it will, then sand back and respray the colour. Next time, run the Step 04 swatch test all the way through the clear before touching the real body.
Grille recess paint bleeds onto the gold rim	Mask right up to the honeycomb opening before brushing the black on, and wipe any bleed off with a cocktail stick while it's still wet.
Steel shot rattles	Use more epoxy, not less, and let it fully gel with the body tipped so the shot settles low and flat before the collar goes on.
Prototype still tips easily with the weight fitted	Add more shot, kept as low and central as the collar allows, and note the extra weight for the ODM.



The knob's boss, glued into the body's own hole, no dowel needed.

REFERENCE

Test record

Write down what you learn before changing anything. It becomes the brief for the industrial designer and ODM.

Total weight fitted			
Steel shot used			
Paint system used			
Fit	☐ Loose	☐ About right	☐ Tight
Finish	☐ Rough	☐ Good	☐ Excellent
Stability, unweighted	☐ Tips easily	☐ Some resistance	☐ Solid
Stability, weighted	☐ Tips easily	☐ Some resistance	☐ Solid
Weight and feel	☐ Too light	☐ About right	☐ Too heavy
Proportions vs. the concept	☐ Off	☐ Close	☐ Spot on

NOTES FOR THE INDUSTRIAL DESIGNER AND ODM

BEFORE YOU FINISH

- ☐ Glue fully cured (overnight or longer)
- ☐ Stands level without rocking
- ☐ Tip test done, both with and without the weight
- ☐ Photos taken, both sets
- ☐ Changes noted for prototype two